

On the Maximum of Stirling Numbers of the Second Kind

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A technique is illustrated for finding an estimate of the Stirling numbers of the second kind, $S_{n,r}$ ($1 \leq r \leq n$), as $n \rightarrow \infty$, for a certain range of values of r . It is shown how the estimation can be found to any given degree of approximation. Finally, the location of the maximum of $S_{n,r}$ ($1 \leq r \leq n$) and the value of the maximum are computed.

1. INTRODUCTION

Let $S_{n,r}$ ($1 \leq r \leq n$) denote the number of distinct partitions of a set of n elements into r non-empty subsets. Clearly, $S_{n,r}$ is also the number of different (i) equivalence relations with r equivalence classes, and (ii) Borel fields with r atoms, on a set of size n .

It is well-known [5, p. 69; 3, p. 22] that

$$(1.1) \quad S_{n,r} = \frac{r^n}{r!} \sum_{k=0}^r (-1)^k \binom{r}{k} \left(1 - \frac{k}{r}\right)^n.$$

In the notation of the calculus of finite differences,

$$S_{n,r} = \frac{\Delta^r O^n}{r!}$$

The behavior of these numbers, specially when $n \rightarrow \infty$, has been studied by various authors. The following is a summary of the known results relevant to the present study.

(a) Dobson [4] and Rennie and Dobson [14] proved by elementary arguments that, for any n , there is a number r_n such that $S_{n,r} > S_{n,r-1}$ for $1 \leq r \leq r_n$, and $S_{n,r} > S_{n,r+1}$ for $r \geq r_n + 1$. Thus $S_{n,r}$ ($1 \leq r \leq n$) have a unique maximum at $r = r_n$. Kanold [9] has derived a weaker version of this result.

(b) By analytic methods, Harper [7] and Lieb [12] showed that $\log S_{n,r}$ ($1 \leq r \leq n$) is a strictly concave function of r ; in fact

$$(1.3) \quad S_{n,k}^2 \geq \frac{k}{k-1} S_{n,k+1} \cdot S_{n,k-1}.$$

(c) Harper [7] established a central limit theorem for $S_{n,r}$ ($1 \leq r \leq n$) as $n \rightarrow \infty$, and showed that

$$(1.4) \quad \lim_{n \rightarrow \infty} \frac{r_n \log n}{n} = 1.$$

The relation (1.4) is also derived jointly by Kanold [10] and Harborth [6].

(d) It is easy to show [13] by elementary inequalities that

$$(1.5) \quad 0 \leq \frac{1}{n \log n} \log S_{n,r} - \left(1 - \frac{r}{n}\right) \frac{\log r}{\log n} = O\left(\frac{1}{\log n}\right).$$

Consequently,

$$\lim_{n \rightarrow \infty} \frac{\log S_{n,r}}{n \log n} = 1 - p \quad \text{or} \quad c$$

according as r varies with n such that

$$\lim_{n \rightarrow \infty} \frac{r}{n} = p \neq 0$$

or

$$\lim_{n \rightarrow \infty} \frac{r}{n} = 0 \quad \text{and} \quad \lim_{n \rightarrow \infty} \frac{\log r}{\log n} = c \quad (0 \leq c \leq 1),$$

respectively.

(e) By elementary methods, it was shown by Rennie and Dobson [14] that

$$r_n = \frac{n}{\log n} + O[n(\log \log n)^{1/2} (\log n)^{-3/2}],$$

$$\max_{1 \leq r \leq n} \log S_{n,r} = n \log n - n \log \log n - n + O\left(n \frac{\log \log n}{\log n}\right),$$

and by the author [13] that

$$(1.6) \quad r_n = \frac{n}{\log n} + O[n(\log n)^{-3/2}],$$

$$(1.7) \quad \max_{1 \leq r \leq n} \log S_{n,r} = n \log n - n \log \log n - n - n \log \left(1 - \frac{\log \log n}{\log n}\right) + O\left(\frac{n}{\log n}\right).$$

(f) Barton *et al.* [1] demonstrated that the most accurate asymptotic estimate known for $S_{n,r}$ is that given by Laplace [11]. They compare the estimates given by Laplace, Cayley [2], DeMoivre, Hsu [8] etc. Laplace's estimate is somewhat complicated and it is not easy to locate the maximum of $S_{n,r}$ ($1 \leq r \leq n$) from it.

(g) For the Bell numbers $f(n) = \sum_{r=1}^n S_{n,r}$, the most accurate asymptotic expansion is that of Binet and Szekeres [15], derived by complex variable techniques. If $\beta e^\beta = n$, they find

$$f(n) = (\beta + 1)^{-1/2} \exp[n(\beta - 1 + \beta^{-1}) - 1] \\ \times \{1 - \beta^2(2\beta^2 + 7\beta + 10)(24n)^{-1}(\beta + 1)^{-3} + O(\beta^2 n^{-2})\};$$

and, for $r < n/\log n$,

$$(1.8) \quad S_{n,r} = \frac{r^n}{r!} \exp \left[\left(\frac{n}{2r} - r \right) e^{-n/r} \right] \left\{ 1 + O \left(\frac{1}{n} \right) \right\}.$$

By an elementary method, it was shown by the author [13] that

$$\frac{1}{n} \log f(n) = \log n - \log \log n - 1 - \log \left(1 - \frac{\log \log n}{\log n} \right) \\ + O \left(\frac{1}{\log n} \right).$$

The object of the present study is to develop an elementary method for estimating $S_{n,r}$ asymptotically to any desired degree of approximation for values of r around the maximum. The sum (1.1) is treated in a natural manner and the results obtained constitute a satisfactory improvement on those previously known. A similar method could be applied to find approximations $S_{n,r}$ for values of r other than those treated here.

In Section 2 it is shown how an estimate of $S_{n,r}$ can be found to any given degree of approximation, for a certain range of values of r around the maximum of $S_{n,r}$. The estimate is used in Section 3 to locate and evaluate the maximum of $S_{n,r}$ (Theorem 2.2 and Theorem 3.1) to a given degree of approximation.

2. APPROXIMATION TO $S_{n,r}$

Let $n \rightarrow \infty$. In the following, the bounds implied by the O and o symbols are absolute constants.

We shall use the Stirling's approximation in the convenient form [5, Chapter 2, problem 27]:

$$(2.1) \quad m! = \sqrt{2\pi} \exp \left[\left(m + \frac{1}{2} \right) \log m - m + \frac{1}{12m} - \frac{1}{360m^3} + \cdots \right].$$

For $k < r$,

$$\begin{aligned} \log \frac{r!}{(r-k)!} &= k \log r - \left(r - k + \frac{1}{2}\right) \log \left(1 - \frac{k}{r}\right) \\ &\quad - k - \frac{k}{12r^2} \left(1 - \frac{k}{r}\right)^{-1} + O\left(\frac{k}{r^4}\right), \end{aligned}$$

so that the k -th summand in (1.1) admits of a natural expansion in powers of k/r . For simplicity, we retain terms upto $1/r^3$ and n/r^4 , and obtain

$$(2.2) \quad \binom{r}{k} \left(1 - \frac{k}{r}\right)^n = \frac{\exp(k\lambda)}{k!} \exp \left[-\frac{k^2}{r^2} \left(\frac{n+r}{2} - \frac{1}{4}\right) - \frac{k^3}{r^2} \left(\frac{n}{3r} + \frac{1}{6}\right) + O\left(\frac{k^4}{r^3} + \frac{nk^4}{r^4}\right) \right]$$

where

$$(2.3) \quad \lambda = \log r - \frac{n}{r} + \frac{1}{2r} - \frac{1}{12r^2}.$$

Such an expansion is the starting point of our analysis. For different ranges of values of r , one can find corresponding estimates of the last exponential on the right side of (2.2), and apply the same technique as below. For efficient estimation by this method, we clearly require the 0-term to be negligible.

A straightforward estimate is obtained by expansion in powers of k/r as follows.

LEMMA 2.1. *Let r and k ($1 \leq k < r \leq$) vary with n so that $nk^2/r^2 \rightarrow 0$. Then, with λ as in (2.3),*

$$(2.4) \quad \binom{r}{k} \left(1 - \frac{k}{r}\right)^n = \frac{\exp(k\lambda)}{k!} \left[1 - \frac{k^2}{r^2} \left(\frac{n+r}{2} - \frac{1}{4}\right) - \frac{k^3}{r^2} \left(\frac{n}{3r} + \frac{1}{6}\right) + \frac{k^4}{2r^4} \left(\frac{n+r}{2}\right)^2 + O\left(\frac{n^3k^6}{r^6}\right) \right]$$

Proof. In (2.3), use the expansion

$$\exp y = 1 + y + \frac{y^2}{2!} + O(y^3)$$

when $|y| < 4$. The terms of higher order than $1/r^2$ and n^2/r^4 are collected together to yield a quantity which is

$$O\left(\frac{k^4}{r^3} + \frac{nk^4}{r^4} + \frac{nk^5}{r^4} + \frac{nk^6}{r^6} + \frac{n^3k^6}{r^6} + \frac{n^3k^8}{r^8}\right) = O\left(\frac{n^3k^6}{r^6}\right).$$

This establishes the lemma. It is in fact sufficient for (2.4) that $nk^2/r^2 < 4$.

From Bonferroni's inequalities (see Feller [5, Chapter 5]) it follows that $S_{n,r}$ may be approximated by the partial sum of the first K terms in (1.1):

(2.5)

$$\left| S_{n,r} - \frac{r^n}{r!} \sum_{k=0}^{R-1} (-1)^k \binom{r}{k} \left(1 - \frac{k}{r}\right)^n \right| < \frac{r^n}{r!} \binom{r}{K} \left(1 - \frac{K}{r}\right)^n, \quad (1 \leq K \leq r).$$

It is easy to verify that, for any fixed j ($1 \leq j \leq K-1$), with $e^\lambda < K-j+1$,

$$(2.6) \quad \left| \sum_{k=0}^{K-1} (-1)^k \frac{\exp(k\lambda)}{k!} (k)_j - (-1)^j e^{j\lambda} \exp(-e^\lambda) \right| < \frac{e^{K\lambda}}{(K-j)!},$$

where

$$(k)_j = k(k-1) \cdots (k-j+1).$$

Observe further that

$$(2.7) \quad \begin{aligned} k^2 &= (k)_2 + k; \\ k^3 &= (k)_3 + 3(k)_2 + k; \\ k^4 &= (k)_4 + 6(k)_3 + 7(k)_2 + k. \end{aligned}$$

It is now clear how (2.5) may be used. We substitute the expansion (2.4) in (2.5) and perform the term-by-term summation by means of (2.7) and (2.6). The error turns out to be

$$\begin{aligned} & \frac{r^n}{r!} \left[O \left\{ \binom{r}{K} \left(1 - \frac{K}{r}\right)^n \right\} + O \left\{ \frac{n^3}{r^6} \left(e^{6\lambda} \exp(-e^\lambda) + \frac{e^{K\lambda}}{(K-6)!} \right) \right\} \right. \\ & \quad \left. + O \left\{ \frac{e^{K\lambda}}{K!} + \frac{n}{2r^2} \frac{e^{K\lambda}}{(K-2)!} + \frac{n}{3r^3} \frac{e^{K\lambda}}{(K-3)!} + \frac{n^2}{r^4} \frac{e^{K\lambda}}{(K-4)!} \right\} \right] \\ & = \frac{r^n}{r!} O \left[\frac{n^3}{r^6} e^{6\lambda} \exp(-e^\lambda) + \frac{e^{K\lambda}}{K!} \right], \end{aligned}$$

provided $e^\lambda < K-6$ and $nK^2/r^2 \rightarrow 0$. The following result is thus proved:

LEMMA 2.2. *Let r and K ($1 \leq K < r \leq n$) vary with n so that*

$$(2.8) \quad \frac{nK^2}{r^2} \rightarrow 0 \quad \text{and} \quad e^\lambda < K-6.$$

Then

(2.9)

$$S_{n,r} = \frac{r^n}{r!} \exp(-e^\lambda) \left[1 - \frac{e^{2\lambda} - e^\lambda}{r^2} \left(\frac{n+r}{2} - \frac{1}{4} \right) - \frac{-e^{3\lambda} + 3e^{2\lambda} - e^\lambda}{r^2} \left(\frac{n}{3r} + \frac{1}{6} \right) + \frac{e^{4\lambda} - 6e^{3\lambda} + 7e^{2\lambda} - e^\lambda}{2r^4} \left(\frac{n+r}{2} \right)^2 + O \left(\frac{n^3 e^{6\lambda}}{r^6} + \exp(e^\lambda) \frac{e^{K\lambda}}{K!} \right) \right].$$

The question now arises as to when the 0-term in (2.9) is negligible. This will be the “natural range” of values of r for which the approximation (2.4) yields an efficient estimate for $S_{n,r}$. The following lemma provides an answer:

LEMMA 2.3. *A choice of K satisfying the conditions*

$$(2.10) \quad \frac{nK^2}{r^2} \rightarrow 0 \quad \text{and} \quad \exp(e^\lambda) \frac{e^{K\lambda}}{K!} \rightarrow 0$$

exists if and only if r varies with n so that

$$(2.11) \quad \frac{r}{\sqrt{n}} \rightarrow \infty \quad \text{and} \quad \frac{n}{r} - \log \sqrt{n} \rightarrow \infty.$$

Moreover, in such a case, K can be chosen so that for any fixed $m > 0$,

$$(2.12) \quad \exp(e^\lambda) \frac{e^{K\lambda}}{K!} = o \left\{ \left(\frac{ne^{2\lambda}}{r^2} \right)^m \right\}.$$

Proof. The condition $nK^2/r^2 \rightarrow 0$ is satisfied if and only if $r/\sqrt{n} \rightarrow \infty$, because if $r/\sqrt{n} = \phi \rightarrow \infty$, one can choose K such that $K/\phi \rightarrow 0$. The second condition in (2.10) provides an upper bound for r as follows. In view of (2.3),

$$(2.13) \quad \exp(e^\lambda) \frac{e^{K\lambda}}{K!} = \exp K \left[\left(\frac{r}{K} e^{-n/r} \right) + \log \left(\frac{r}{K} e^{-n/r} \right) + 1 - \frac{1}{2} \frac{\log K}{K} + O \left(\frac{1}{K} \right) \right],$$

which is vanishing only if $(r/K) e^{-n/r} < e^{-1}$, which implies that $\sqrt{n} K/r > e \sqrt{n} e^{-n/r}$, whence $\sqrt{n} e^{-n/r} \rightarrow 0$. Conversely, let

$$(n/r) - \log \sqrt{n} \rightarrow \infty.$$

In case $re^{-n/r}$ is bounded, it is enough to choose $K \rightarrow \infty$ such that $\sqrt{n} K/r \rightarrow 0$; in case $re^{-n/r} \rightarrow \infty$, we choose K such that $(r/K) e^{-n/r} \sim e^{-2}$ and verify that $K \rightarrow \infty, \sqrt{n} K/r \rightarrow 0$; in either case, the quantity on the right side of (2.13) tends to zero.

It remains to prove (2.12). One may assume that $K \rightarrow \infty$ and $(r/K) e^{-n/r} < e^{-1}$, or, $e^\lambda < e^{-1}K$, in view of the above. The ratio of the two quantities in (2.12) may be written as

$$(2.14) \quad \exp K \left[\left(\frac{r}{K} e^{-n/r} \right) + \left(1 - \frac{2m}{K} \right) \log \left(\frac{r}{K} e^{-n/r} \right) + 1 \right. \\ \left. + \frac{m}{K} \log \frac{r^2}{n} + O \left(\frac{\log K}{K} \right) \right].$$

Write $r/\sqrt{n} = \phi$, and choose K satisfying the additional conditions $K/\phi \rightarrow 0$ and $(2m/K) \log \phi \rightarrow 0$. Then the difference between the quantities inside the square brackets of (2.13) and (2.14) is vanishingly small, and the proof used in the previous paragraph can be used to establish the required fact.

In view of the above lemma, the second quantity inside the 0-term of (2.9) can be ignored. Thus the "dummy" variable K disappears and we may write down a general result as follows. Let

$$(2.15) \quad k^m = \sum_{j=1}^{m-1} \alpha_{mj}(k)_j$$

as in (2.7), where α 's are suitable constants, and suppose the expansion in (2.4) is continued to m terms

$$(2.16) \quad \left(\frac{r}{k} \right) \left(1 - \frac{k}{r} \right)^n = \frac{\exp(k\lambda)}{k!} \left[1 + \sum_{i=2}^{2m-1} a_i k^i + O \left(\frac{nk^2}{r^2} \right)^m \right],$$

where a_i are rational functions of n and r .

THEOREM 2.1. *In the notation of (2.15) and (2.16), if r varies with n such that $r/\sqrt{n} \rightarrow \infty$ and $n/r - \log \sqrt{n} \rightarrow \infty$, we may write, with λ as in (2.3),*

$$S_{n,r} = \frac{r^n}{r!} \exp(-e^\lambda) \left[1 + \sum_{i=2}^{2m-1} a_i \left(\sum_{j=1}^{i-1} (-1)^j \alpha_{ij} e^{j\lambda} \right) + O \left(\frac{ne^{2\lambda}}{r^2} \right)^m \right].$$

In particular, with $m = 3$, the expansion (2.9) is valid with the 0-term replaced by $O(n^3 e^{6\lambda}/r^6)$.

We now substitute for λ from (2.3) and simplify (2.9) to the following:

THEOREM 2.2. *Let r vary with n so that*

$$r/\sqrt{n} \rightarrow \infty \quad \text{and} \quad (n/r) - \log \sqrt{n} \rightarrow \infty.$$

Let

$$\lambda = \log r - \frac{n}{r} + \frac{1}{2r} - \frac{1}{12r^2}.$$

Then

(2.17)

$$\begin{aligned} S_{n,r} = & \frac{r^n}{r!} \exp(-e^\lambda) \left[1 + \frac{1}{r} e^{-n/r} \left(\frac{n+r}{2} - \frac{n^2}{8r^2} + \frac{n}{3r} + \frac{1}{24} \right) \right. \\ & - e^{-2n/r} \left(\frac{n+r}{2} - \frac{7n^2}{8r^2} - \frac{n}{4r} - \frac{1}{8} \right) - re^{-3n/r} \left(\frac{3n^2}{4r^2} + \frac{7n}{6r} + \frac{7}{12} \right) \\ & \left. + e^{-4n/r} \frac{(n+r)^2}{8} + R_1 \right], \end{aligned}$$

where

$$\begin{aligned} R_1 = & O \left(\frac{n^2}{r^4} e^{-n/r} + \frac{n^2}{r^3} e^{-2n/r} + \frac{n^2}{r^2} e^{-3n/r} + \frac{n^2}{r} e^{-4n/r} + n^3 e^{-6n/r} \right). \\ = & \begin{cases} O \left(\frac{n^2}{r^4} e^{-n/r} + n^3 e^{-6n/r} \right), & \text{if } re^{-n/r} \leq 1, \\ O(n^3 e^{-6n/r}), & \text{if } re^{-n/r} > 1. \end{cases} \end{aligned}$$

COROLLARY. *Under the same conditions on r as in the Theorem 2.2,*

(2.18)

$$\begin{aligned} S_{n,r} = & \frac{r^n}{r!} \exp \left[\left(\frac{n}{2r} - r \right) e^{-n/r} - \frac{1}{r} e^{-n/r} \left(\frac{n^2}{8r^2} - \frac{n}{3r} \right) \right. \\ & \left. - e^{-2n/r} \left(\frac{n+r}{2} - \frac{3n^2}{4r^2} \right) - re^{-3n/r} \left(\frac{n^2}{2r^2} + \frac{2n}{3r} + \frac{1}{3} \right) + R_2 \right], \end{aligned}$$

where

$$\begin{aligned} R_2 = & O \left(\frac{n^2}{r^4} e^{-n/r} + \frac{n^3}{r^4} e^{-2n/r} + \frac{n^3}{r^3} e^{-3n/r} + \frac{n^3}{r^2} e^{-4n/r} + \frac{n^3}{r} e^{-5n/r} \right. \\ & \left. + n^3 e^{-6n/r} \right), \\ = & \begin{cases} O \left(\frac{n^2}{r^4} e^{-n/r} + \frac{n^3}{r^4} e^{-2n/r} \right), & \text{if } re^{-n/r} \leq 1, \\ O(n^3 e^{-6n/r}), & \text{if } re^{-n/r} > 1. \end{cases} \end{aligned}$$

Proof. In (2.17), expand $\log(r! \exp(e^\lambda) S_{n,r}/r^n)$ by means of the relation

$$\log(1 + x) = x - \frac{x^2}{2} + O(x^3),$$

when $x = o(1)$, and simplify to obtain (2.18).

The maximum of $S_{n,r}$ occurs within the range of values of r for which the expansion (2.18) holds. We shall see in the next section that the maximum occurs at a value of r such that $n/r = \log n - \log \log n + o(1)$. In fact, the maximum occurs near the root of the equation $re^{-n/r} = 1$. It is clear that, for such values of r , (2.17) and (2.18) provide an expansion of $S_{n,r}$ in powers of $1/n$, and the 0-term is of the form

$$O\left(\frac{(\log n)^6}{n^3}\right).$$

3. THE MAXIMUM OF $S_{n,r}$

Because of the concavity of $\log S_{n,r}$, it is enough to locate a maximum. Alternatively, one could prove by means of elementary inequalities that $S_{n,r}$ assumes smaller values outside the range specified by Theorem 2.2.

Introducing the variable p by the relation

$$(3.1) \quad p = r/n,$$

and using the Stirling's approximation (2.3), we may write (2.18) as follows:

$$(3.2) \quad (1/n) \log(\sqrt{2\pi} S_{n,r}) = f(p) + g(p) + R,$$

where

$$(3.3) \quad f(p) = (1 - p) \log np + p,$$

$$(3.4) \quad g(p) = -\frac{\log np}{2n} - \frac{p^{-1}}{12n^2} - pe^{-p^{-1}} + e^{-p^{-1}} \left(\frac{p^{-1}}{2n} - \frac{p^{-3}}{8n^2} + \frac{p^{-2}}{3n^2} \right) \\ - e^{-2p^{-1}} \left(\frac{1+p}{2} - \frac{3p^{-2}}{4n} \right) - e^{-3p^{-1}} \left(\frac{p^{-1}}{2} + \frac{2}{3} + \frac{p}{3} \right),$$

and

$$(3.5) \quad R = R_2/n.$$

The maximum of f is easy to determine.

LEMMA 3.1. $f(p)$, $0 < p < 1$, is a strictly concave function of p and its maximum occurs when $p^{-1} = \beta$ where β is the root of the equation

$$(3.6) \quad \beta e^\beta = n.$$

Moreover,

$$(3.7) \quad \max_{0 < p < 1} f(p) = \beta - 1 + \beta^{-1},$$

and

$$(3.8) \quad \beta = \log n - \log \log n - \log \left(1 - \frac{\log \log n + o(1)}{\log n} \right).$$

Proof. All the assertions except (3.8) follow from the differential coefficients

$$(3.9) \quad f'(p) = p^{-1} + \log p^{-1} - \log n,$$

$$(3.10) \quad f''(p) = -p^{-2} - p^{-1}.$$

If we write $\beta = \log n - \log \log n + \delta$, it follows that

$$(3.11) \quad \delta = -\log \left(1 - \frac{\log \log n - \delta}{\log n} \right),$$

so that $\delta = o(1)$. This completes the proof of the lemma.

Clearly,

$$\delta = \frac{\log \log n}{\log n} (1 + o(1)),$$

so that from (3.11), a simple expansion yields

$$(3.12) \quad \delta \left(1 + \frac{1}{\log n} \right) = a + \frac{1}{2} a^2 + \frac{1}{3} a^3 (1 + o(1)),$$

where

$$a = \frac{\log \log n}{\log n}.$$

It is easy to verify that

$$(3.13) \quad \begin{aligned} g'(p) = & -\frac{p^{-1}}{2n} - e^{-p^{-1}}(1 + p^{-1}) + \frac{p^{-2}}{12n^2} + e^{-p^{-1}} \\ & \times \left(\frac{p^{-3} - p^{-2}}{2n} - \frac{3p^{-5} - 17p^{-4} + 16p^{-3}}{24n^2} \right) \\ & - e^{-2p^{-1}} \left\{ p^{-2} + p^{-1} + \frac{1}{2} - 3 \cdot \frac{p^{-4} - p^{-3}}{2n} \right\} \\ & - e^{-3p^{-1}} \left(\frac{3}{2} p^{-3} + \frac{3}{2} p^{-2} + p^{-1} + \frac{1}{3} \right). \end{aligned}$$

In order to find the maximum of $f(p) + g(p)$ we are going to proceed as follows: When $p^{-1} \sim \beta$, we find that $g'(p) \sim -\beta^2/n$; it follows that the root of the equation $f'(p) + g'(p) = 0$ occurs "near" the root β^{-1} of the equation $f'(p) = 0$, since $f'(p)$ is continuously differentiable at $p = \beta^{-1}$. We first estimate the "correction" h needed in β in order to satisfy the former equation. It turns out that $h \sim \beta^2/n$ so that $f(p) + g(p)$ can be evaluated at $p = (\beta + h)^{-1}$ by means of a Taylor expansion around β^{-1} . The two equations $f'(p) = 0$ and $f'(p) + g'(p) = 0$ cannot be solved exactly, but the solution to the latter may be found to any degree of approximation in a series of powers of $1/n$. The details of the procedure are given below.

When

$$(3.14) \quad p^{-1} = \beta + h, \quad h = o(\beta),$$

we find that

$$(3.15) \quad \begin{aligned} f'(p) &= \beta + h + \log(\beta + h) - \log n \\ &= h(1 + 1/\beta) - h^2/2\beta^2 + h^3/3\beta^3 + O(h^4/\beta^4), \end{aligned}$$

and

$$(3.16) \quad \begin{aligned} g'(p) &= -\frac{\beta}{n} \left(\beta + \frac{3}{2} \right) - \frac{\beta^2}{2n^2} \left(\beta^2 + 3\beta + \frac{5}{6} \right) \\ &\quad - \frac{\beta^3}{24n^3} (3\beta^3 + 55\beta^2 + 40\beta + 8) \\ &\quad + \frac{h}{n} \left(\beta^2 - \frac{1}{2} \right) + \frac{h}{n^2} \left(\frac{3}{2} \beta^4 + 2\beta^3 - \beta^2 + \frac{\beta}{6} \right) - \frac{h^2}{2n} \beta(\beta - 1) \\ &\quad + O \left(\frac{\beta^6 h}{n^3} + \frac{\beta^4 h^2}{n^2} + \frac{\beta^2 h^3}{n} \right). \end{aligned}$$

The equation $f'(p) = -g'(p)$ is clearly satisfied when

$$(3.17) \quad h = \frac{1}{n} \frac{\beta^2(\beta + \frac{3}{2})}{\beta + 1} + \frac{1}{n^2} \frac{\beta^3}{2(\beta + 1)^3} \left(-\beta^4 + \frac{41}{6} \beta^2 + \frac{61}{6} \beta + \frac{55}{12} \right) + \frac{1}{n^3} O(\beta^6)$$

In fact, if necessary, the expansion (3.17) may be easily continued further.

Now

$$(3.18) \quad (\beta + h)^{-1} = \beta^{-1} - \frac{h}{\beta^2} \left(1 + \frac{h}{\beta} \right)^{-1},$$

and

$$(3.19) \quad h = \frac{\beta^2}{n} (1 + o(1)), \quad (\beta + h)^{-1} \sim \beta^{-1},$$

so that the Taylor's expansion yields, after some simplification,

(3.20)

$$f((\beta + h)^{-1}) = \beta - 1 + \beta^{-1} - \frac{h^2}{2\beta^3} (\beta + 1) + \frac{h^3}{3\beta^4} (2\beta + \frac{5}{2}) + O\left(\frac{h^4}{\beta^4}\right)$$

and

$$(3.21) \quad \begin{aligned} g((\beta + h)^{-1}) = & -\frac{\beta + 2}{2n} - \frac{7\beta}{12n^2} + \frac{\beta^2}{n^3} \left(\frac{\beta^2}{8} - \frac{\beta}{3} - \frac{1}{3} \right) \\ & + \frac{h}{n} \frac{(\beta + 3/2)}{\beta} + \frac{h}{n^2} \left(\frac{1}{2} \beta (\beta + 3) + \frac{5}{12} \right) \\ & - \frac{h^2}{n\beta^2} \left(\frac{\beta^2}{2} + \beta + \frac{5}{4} \right) + O\left(\frac{\beta^6}{n^4}\right). \end{aligned}$$

However, from (3.15) and (3.16), the equation for h yields

$$\begin{aligned} \beta^2 \left[\frac{h}{n} \frac{\beta + 3/2}{\beta} + \frac{h}{2n^2} \left(\beta^2 + 3\beta + \frac{5}{6} \right) \right] &= h^2 \frac{\beta + 1}{\beta} - \frac{h^3}{2\beta^2} + \frac{h^2}{n} \left(\beta^2 - \frac{1}{2} \right) \\ &+ O\left(\frac{\beta^8}{n^4}\right), \end{aligned}$$

so that, in (3.21), we may replace the h -terms by those involving h^2 and h^3 , and obtain finally

$$\begin{aligned} f((\beta + h)^{-1}) + g((\beta + h)^{-1}) &= \beta - 1 + \beta^{-1} - \frac{\beta + 2}{2n} - \frac{7\beta}{12n^2} + \frac{\beta^2}{n^3} \left(\frac{\beta^2}{8} - \frac{\beta}{3} - \frac{1}{3} \right) \\ &+ \frac{h^2}{2\beta^3} (\beta + 1) + \frac{h^3}{3\beta^4} (2\beta + 1) + \frac{h^2}{n\beta^2} \left(\frac{\beta^2}{2} - \beta - \frac{7}{4} \right) + O\left(\frac{\beta^6}{n^4}\right). \end{aligned}$$

It only remains to substitute for h from (3.17). Substituting and simplifying, we obtain the maximum of $f(p) + g(p)$. When $p^{-1} \sim \beta$, it is easy to see that $R_2 = O(\beta^6/n^3)$. We may therefore, write the result as follows:

THEOREM 3.1. *The maximum of $S_{n,r}$ occurs when r satisfies the relation*

$$\frac{n}{r} = \beta + \frac{1}{n} \frac{\beta^2(\beta + 3/2)}{\beta + 1} + \frac{1}{n^2} \frac{\beta^3}{2(\beta + 1)^3} \left(-\beta^4 + \frac{41}{6}\beta^3 + \frac{61}{6}\beta + \frac{55}{12} \right) + O\left(\frac{\beta^6}{n^3}\right),$$

where β is the solution of the equation

$$\beta e^\beta = n.$$

Moreover,

$$\begin{aligned} \max_r \frac{1}{n} \log(S_{n,r} \sqrt{2\pi}) \\ = \beta - 1 + \beta^{-1} - \frac{\beta + 2}{2n} + \frac{\beta^2}{24n^2(\beta + 1)} (12\beta^2 + 22\beta + 13) \\ + \frac{\beta^2}{24n^3(\beta + 1)^2} \left\{ (\beta + 1)^2 (16\beta^4 + 80\beta^3 + 147\beta^2 + 100\beta + 19) \right. \\ \left. + \left(\beta + \frac{3}{2} \right) \left(12\beta^3 - 5\beta^2 - 58\beta - \frac{79}{2} \right) \right\} + O(\beta^6/n^4) \end{aligned}$$

It may be possible to estimate $f(n) = \sum_{r=1}^n S_{n,r}$ using the above estimates (Theorems 3.1 and 2.2), but this problem will not be examined here.

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